

R Programming & Syntax

SOCI 3238 Digital Sociology · Tutorial 1

Sep 28, 2026 (Mon) 12:30–14:00 · Sino Building 408B

[Your Name] (TA)

Agenda

1. Why R for digital sociology?
2. RStudio & R scripts: the basics
3. Objects, vectors, and data frames
4. Control flow & functions
5. Hands-on: from data to a plot

Why R?

- Open source and **free** — runs everywhere
- Purpose-built for **statistics and data analysis**
- Packages for digital sociology: `dplyr`, `ggplot2`, `igraph`, `quanteda`, `httr` ...
- Great for **reproducible research** (scripts > clicks)

RStudio basics

The four panes

1. Source (scripts)
2. Console
3. Environment
4. Files / Plots / Help

Run the current line with ⌘ + Enter

Tips

- Create an **R Project** per course

Objects & vectors

```
# everything is an object; <- is assignment  
x <- c(2, 4, 6, 8, 10)
```

```
length(x)          # 5  
mean(x)            # 6  
x[x > 5]           # 6 8 10
```

```
# character and logical vectors  
majors <- c("soci", "econ", "soci")  
is_soci <- majors == "soci"  
table(majors)       # counts by value
```

Data frames

```
library(dplyr)

users <- read_csv("data/users.csv")

users |>
  filter(age >= 18) |>
  group_by(education) |>
  summarise(
    n = n(),
    avg_income = mean(income, na.rm = TRUE)
  )
```

“ **Key idea:** a data frame is a spreadsheet. Rows are observations, columns are variables. ”

Control flow & functions

```
# a tiny function
scale01 <- function(x) {
  (x - min(x, na.rm = TRUE)) /
    (max(x, na.rm = TRUE) - min(x, na.rm = TRUE))
}

users$age_std <- scale01(users$age)

# conditionals and loops
for (year in 2020:2026) {
  if (year %% 4 == 0) cat(year, "is a leap year\n")
}
```

Hands-on

Data → table → plot (15 min)

Hands-on: from data to a plot

1. Open `week1/hands-on.R` from the course repo
2. Load the data and inspect it: `glimpse()`, `summary()`
3. Answer:
 - How many observations? Any missing values?
 - What is the mean of your assigned variable by group?
4. Plot the group means with `ggplot2`

```
ggplot(df, aes(group, value)) +  
  geom_boxplot()
```

Before next tutorial

- **Do:** the hands-on exercise end-to-end on your own machine
- **Install:** `tidyverse`, `igraph`, `tidygraph` — needed in Tutorial 2 (network analysis)
- **Ask:** post questions on the course forum — no question is too basic!

Thanks!

Slides & materials: course repo

Questions → course forum / email